



Group Policy Objects (GPOs) in Microsoft Entra ID

Enforcing Security Policy in an Azure AD Environment

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Platform: Microsoft Azure | Tools: GPMC, Entra ID, AD DS

Environment: Windows Server 2022 DC + Windows 11 Pro Client

#GPO

#EntraID

#ActiveDirectory

#AzureLab

#CyberSecurity

#IdentitySecurity

About this Lab Report

This portfolio lab report demonstrates the configuration of Group Policy Objects (GPOs) in a Microsoft Azure-hosted Active Directory environment (lab.roshi). Three enterprise security policies are implemented and verified: Desktop Wallpaper enforcement, Control Panel restriction, and Task Manager disablement — all linked to an HR Organizational Unit and confirmed on a domain-joined Windows 11 Pro client (lab.roshi\mbell).

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1. Executive Summary & GPO Overview

Group Policy Objects (GPOs) are one of the most powerful configuration management and security enforcement mechanisms in Microsoft Active Directory environments. A GPO is a collection of settings that controls the working environment of user and computer accounts. Administrators use GPOs to enforce security baselines, restrict access to sensitive system components, standardize desktop environments, and automate compliance across endpoints — without requiring manual intervention on each machine.

Why GPOs Matter in Enterprise Security

- **Centralized Control:** A single GPO linked to an OU instantly applies to every user or computer in scope — eliminating configuration drift.
- **Least Privilege Enforcement:** GPOs restrict what users can access (Control Panel, Task Manager, Registry) without changing account permissions.
- **Compliance Automation:** GPOs enforce DISA STIG, CIS Benchmarks, and NIST 800-53 controls at scale, directly supporting audit requirements.
- **Incident Response:** During an active incident, GPOs can push lockdown policies — disabling USB drives or blocking internet — across all affected systems simultaneously.
- **Identity Security Layer:** Combined with Entra ID, GPOs add an on-premises policy enforcement layer that complements cloud Conditional Access Policies.

GPOs in the SOC Context

From a SOC perspective, GPOs are a preventive and detective control. Disabling Task Manager prevents malware or malicious insiders from terminating security agent processes (EDR/AV). Restricting Control Panel prevents unauthorized changes to network settings, firewall configurations, and user accounts. Enforcing desktop wallpapers with legal banners satisfies admissibility notice requirements in regulated industries. SOC analysts monitor GPO changes via Windows Event ID 5136 (directory service object modification) as unauthorized GPO modifications are a known persistence technique — MITRE ATT&CK; T1484 (Group Policy Modification).

MITRE ATT&CK Ref**T1484.001 – Group Policy Modification**

Adversaries abuse GPOs to deploy malware, establish persistence, or escalate privileges.

2. Lab Environment Setup

This lab is hosted entirely in Microsoft Azure, simulating an enterprise on-premises Active Directory environment in the cloud. Two virtual machines are connected within a dedicated virtual network and subnet.

Role	OS	VM Name	IP / Notes
Domain Controller	Windows Server 2022 Datacenter Azure Edition	DC-RoshiLab	Static IP, DNS Server AD DS, GPMC installed
Client Workstation	Windows 11 Pro	Client1	DHCP, DNS → DC IP Domain: lab.roshi

Network & AD Configuration

- Resource Group: **rg-roshi** | VNet Subnet: **192.168.100.0/24**
- NSG Inbound Rule: RDP (TCP 3389) restricted to administrator's public IP only
- Domain Name: **lab.roshi** | DNS: all VMs point to DC private IP
- Organizational Unit (OU): **HR** — Client1 and user Max Bell (mbell) placed here
- Shared company wallpaper folder: **\\DC01\\Company\\logo.png**

GPO 1 of 3 — Desktop Wallpaper Enforcement Policy

3. GPO 1 – Desktop Wallpaper Policy

This policy enforces a company-branded desktop wallpaper on all workstations in the HR OU. It prevents users from changing the wallpaper, ensuring corporate branding consistency and — in regulated environments — displaying legal notice banners that satisfy compliance requirements (PCI-DSS, HIPAA acceptable use notice).

1 Open GPMC → Group Policy Objects → Right-click → New

- Navigate: **Server Manager** → **Tools** → **Group Policy Management**
- Expand: Forest → Domains → lab.roshi → **Group Policy Objects**
- Right-click **Group Policy Objects** → select **New**
- Name the new GPO: **Company Wallpaper**

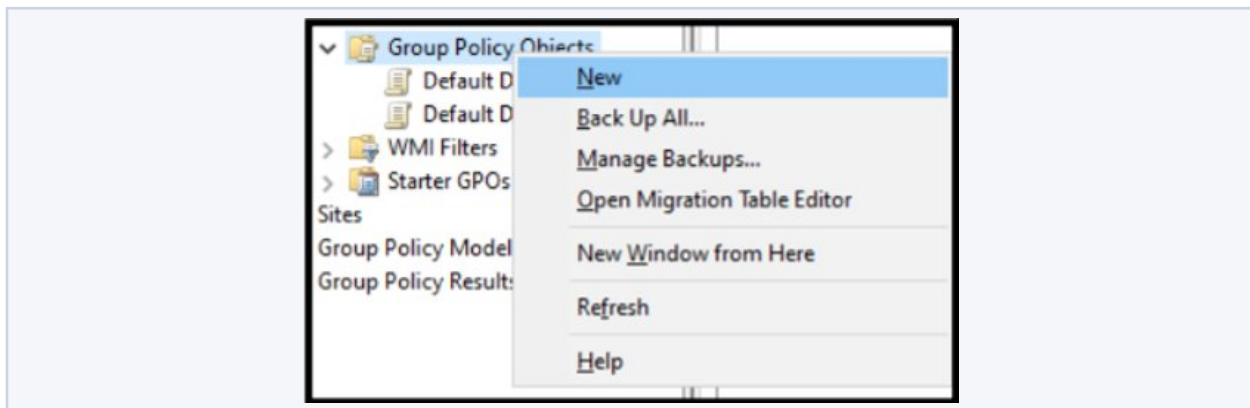


Figure 1 – Right-clicking Group Policy Objects container to create a new GPO

2 Navigate to Desktop Wallpaper setting in GPO Editor

- Right-click the new GPO → **Edit**
- Navigate: User Configuration → Policies → Administrative Templates → Desktop → **Desktop**

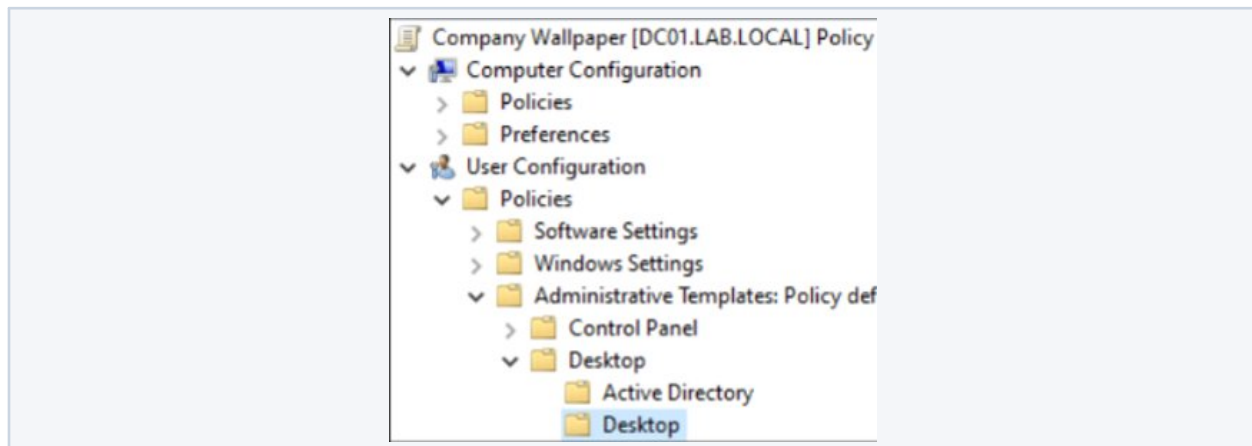


Figure 2 – GPO Editor tree: User Configuration → Administrative Templates → Desktop → Desktop

3 Select Desktop Wallpaper from the settings list

- On the right panel, locate and double-click: **Desktop Wallpaper**

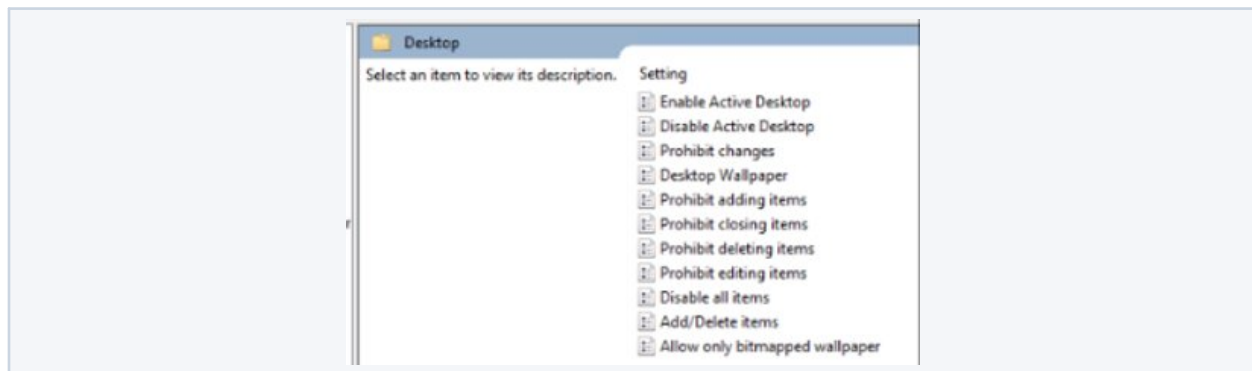


Figure 3 – Desktop settings list showing Desktop Wallpaper option

4 Configure the wallpaper UNC path and enable the policy

- Set radio button to: **Enabled**
- Wallpaper Name (UNC path): **\\DC01\\Company\\logo.png**
- Wallpaper Style: **Fill** → Click **Apply** → **OK**

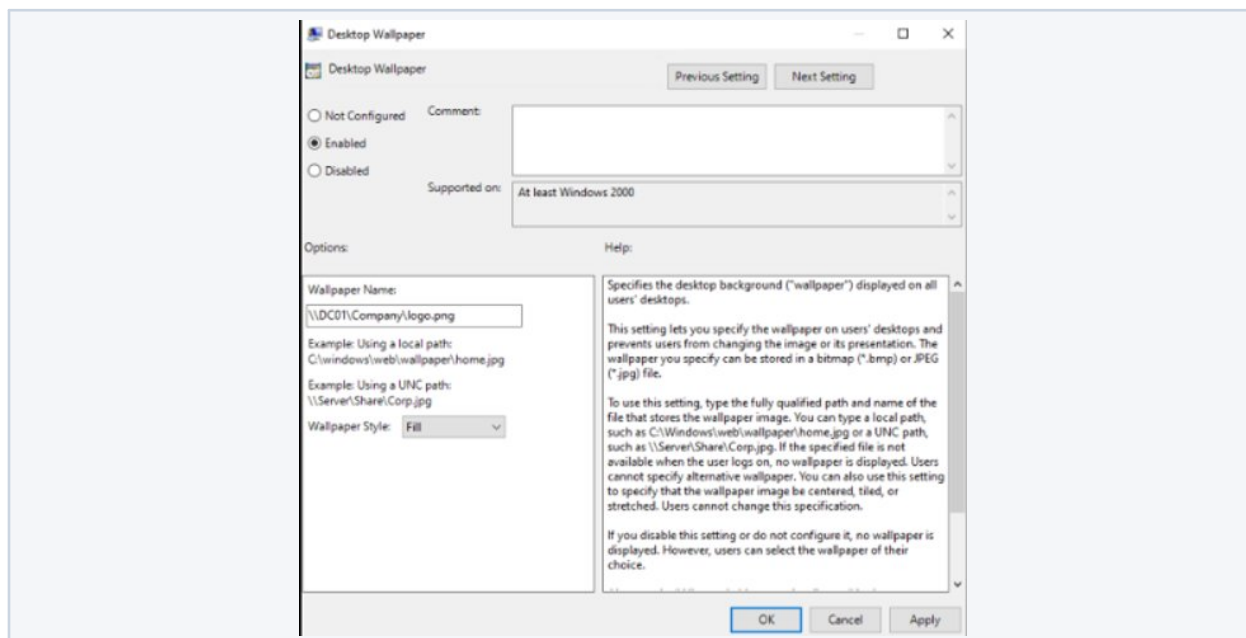


Figure 4 – Desktop Wallpaper policy Enabled with UNC path \\DC01\\Company\\logo.png

GPO 2 of 3 — Control Panel Access Restriction Policy

4. GPO 2 – Disable Control Panel

Restricting Control Panel access is a fundamental least-privilege control. Users with access to Control Panel can modify network adapters, disable Windows Defender, uninstall software, and alter system settings — all of which are security risks in a managed enterprise environment. This GPO removes that access entirely for standard users in the HR OU.

1 Create a new GPO named 'Disable Control Panel'

- In GPMC: Right-click **Group Policy Objects** → **New**
- Name: **Disable Control Panel** → Click OK

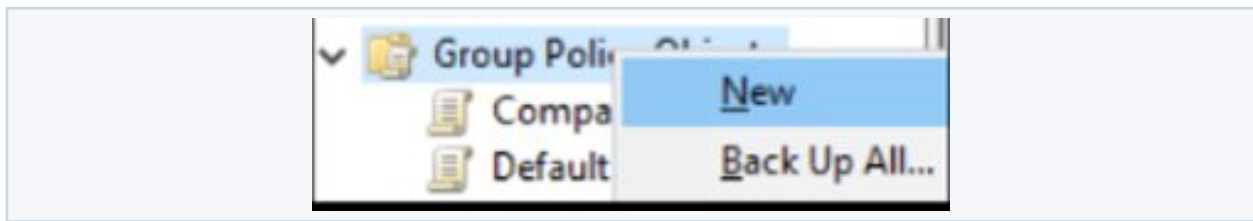


Figure 5 – Creating a new GPO via right-click context menu

2 Right-click the new GPO → Edit, then navigate to Control Panel

- Right-click **Disable Control Panel** GPO → **Edit**

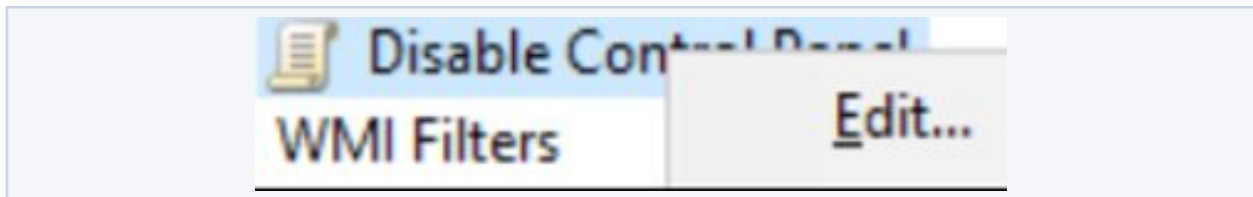


Figure 6 – Right-clicking the Disable Control Panel GPO to open Edit

- Navigate: User Configuration → Policies → Administrative Templates → **Control Panel**

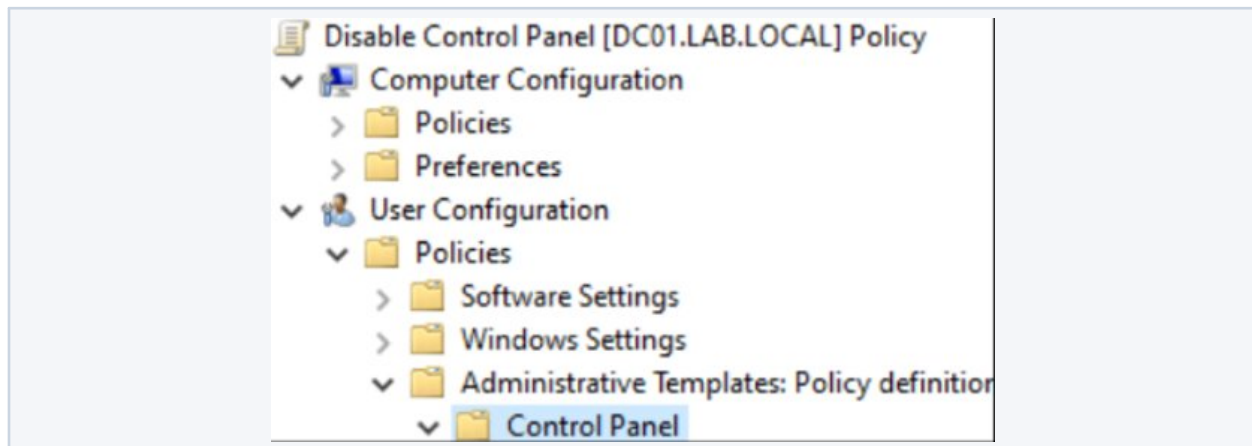


Figure 7 – GPO Editor showing Disable Control Panel [DC01.LAB.LOCAL] Policy tree

3 Select and enable the prohibition policy

- On right panel, locate: **Prohibit access to Control Panel and PC settings**

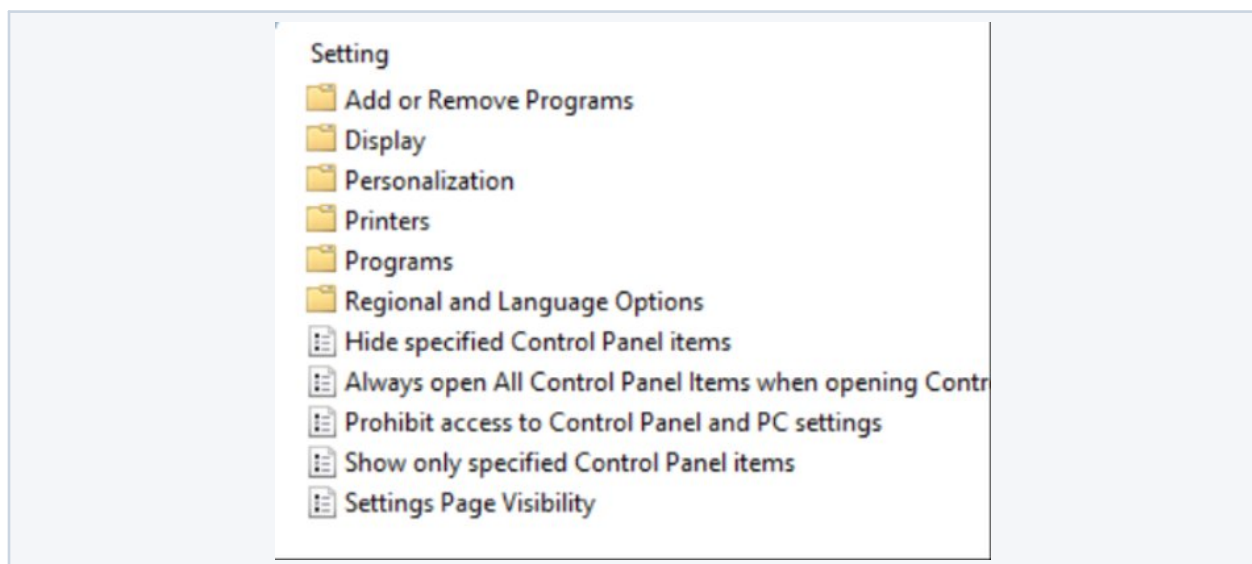


Figure 8 – Control Panel settings list showing 'Prohibit access to Control Panel and PC settings'

- Double-click it → Set to **Enabled** → Click **Apply** → **OK**

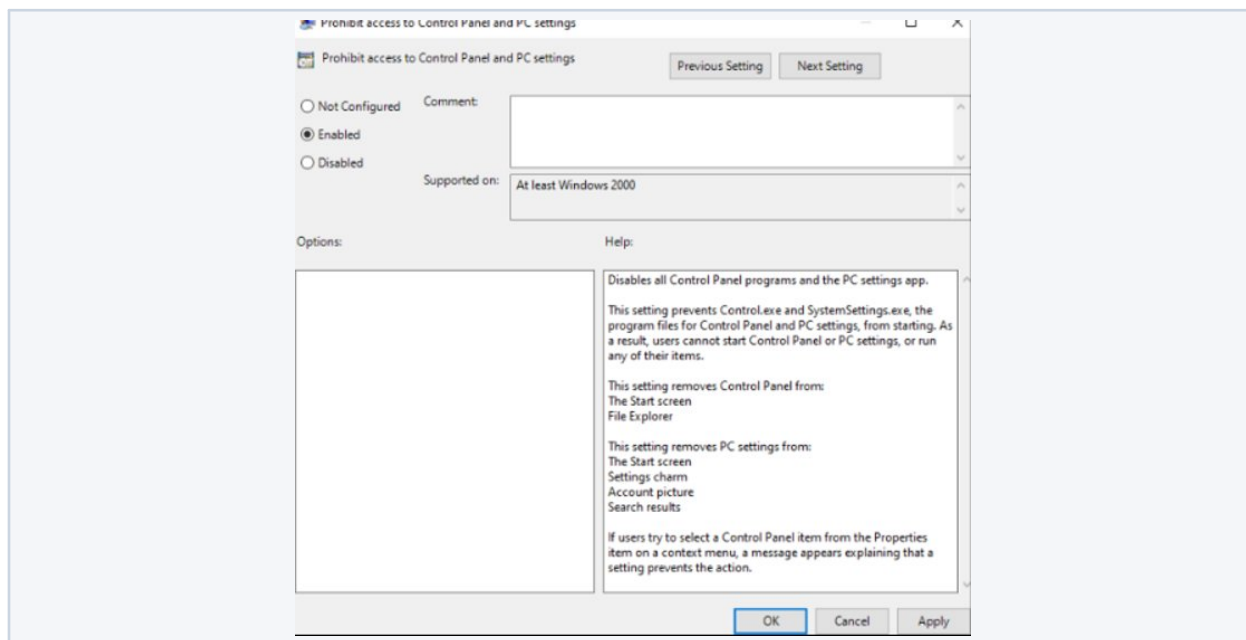


Figure 9 – 'Prohibit access to Control Panel and PC settings' set to Enabled

■ Security Note

This blocks both the classic Control Panel AND the modern Windows Settings app. It prevents users from accessing either.

GPO 3 of 3 — Task Manager Disablement Policy

5. GPO 3 – Disable Task Manager

Disabling Task Manager is an important endpoint hardening control. It prevents users from terminating system processes, killing security agent processes (EDR/AV), or viewing process trees that could reveal monitoring tools. From a SOC perspective, monitoring for attempts to re-enable Task Manager via registry changes is a useful detection signal.

1 Locate all GPOs — confirm all three are created

- In GPMC under Group Policy Objects, all three GPOs should now be visible
- Right-click **Disable Task Manager** → **Edit**

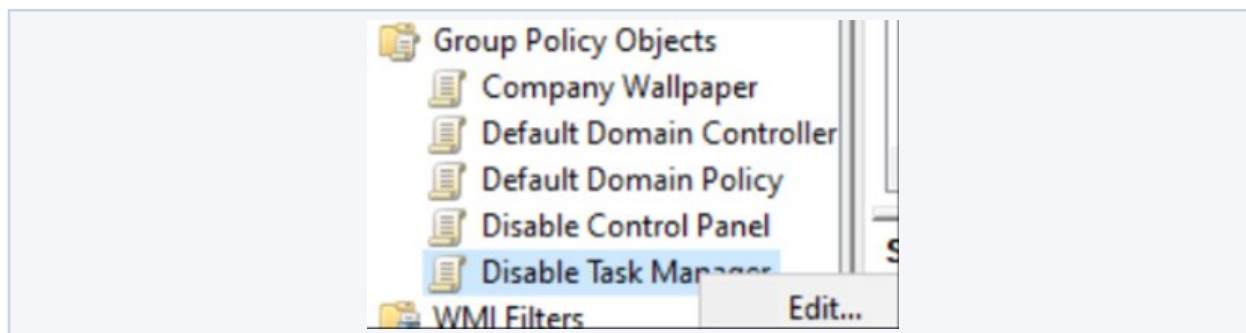


Figure 10 – Group Policy Objects showing Company Wallpaper, Disable Control Panel, and Disable Task Manager

2 Navigate to Ctrl+Alt+Del Options under System

- Navigate: User Configuration → Policies → Administrative Templates → System → **Ctrl+Alt+Del Options**

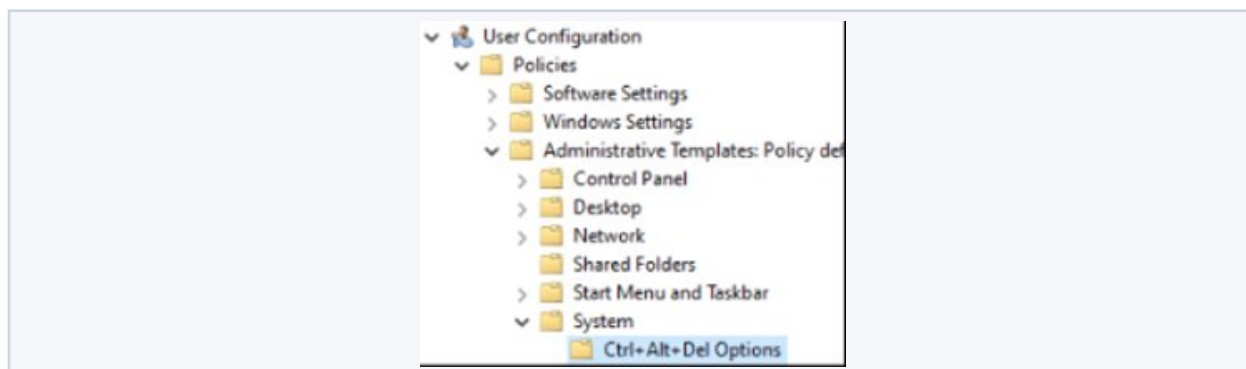


Figure 11 – GPO Editor tree: User Configuration → System → Ctrl+Alt+Del Options

3 Select Remove Task Manager from the settings list

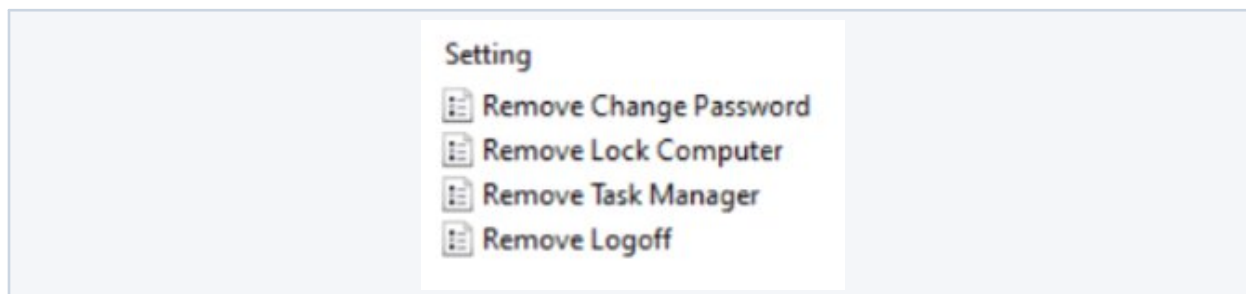


Figure 12 – Ctrl+Alt+Del Options settings list showing Remove Task Manager

4 Enable the Remove Task Manager policy

- Double-click **Remove Task Manager** → Set to **Enabled** → Click **Apply** → **OK**

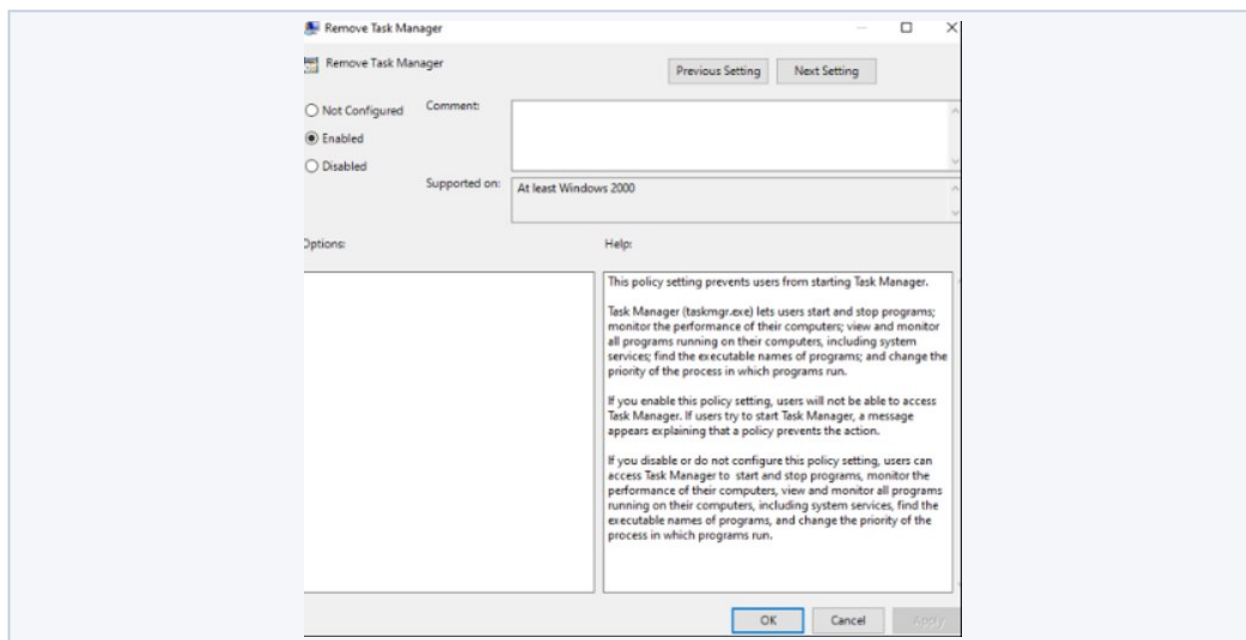


Figure 13 – Remove Task Manager policy set to Enabled

■ **Registry Reference** This GPO writes: HKCU\Software\Microsoft\Windows\CurrentVersion\Policies\System\DisableTaskMgr = 1 SO

6. Linking & Enforcing GPOs on the HR OU

Creating a GPO does not automatically apply it. GPOs must be explicitly linked to a domain, site, or Organizational Unit (OU) to take effect. Setting a GPO to 'Enforced' ensures it cannot be overridden by GPOs lower in the hierarchy — critical in environments where child OUs or local policies might otherwise conflict.

1 Right-click the HR OU → Link an Existing GPO

- In GPMC: Right-click the **HR OU** → **Link an Existing GPO...**

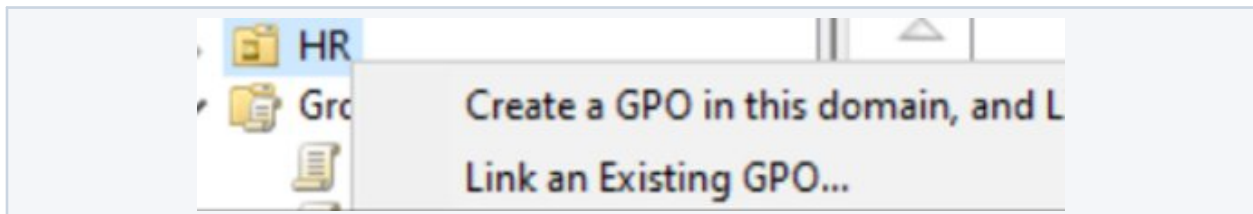


Figure 14 – Right-clicking HR OU to access 'Link an Existing GPO' option

2 Select all three GPOs to link

- In the Select GPO dialog, select: **Company Wallpaper**, **Disable control panel**, **Disable Task Manager**
- Domain: **lab.roshi** — confirm all three appear under Group Policy objects

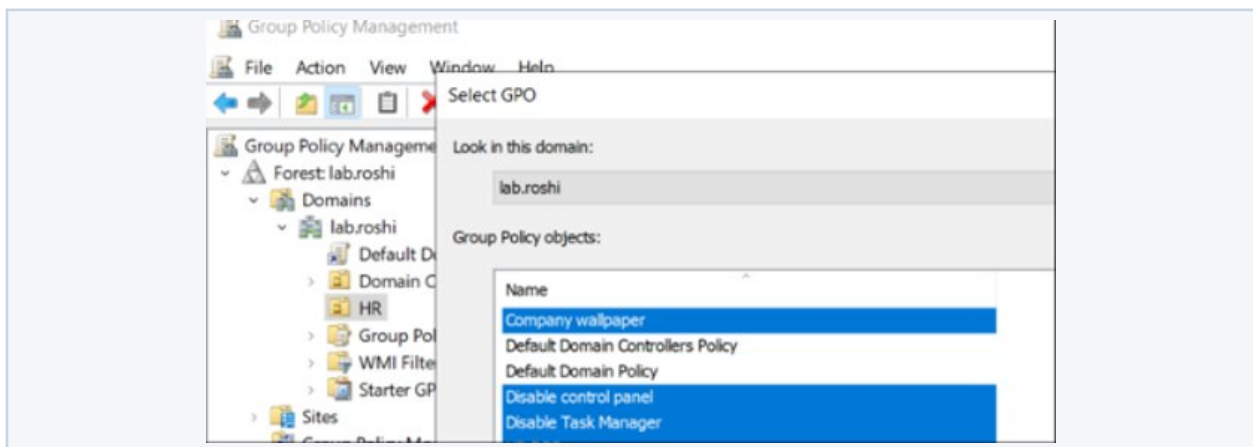


Figure 15 – Select GPO dialog showing all three GPOs selected for linking to HR OU

3 View linked GPOs under HR OU and enforce each

- HR OU now shows all three GPOs in the Linked Group Policy Objects tab
- Right-click each GPO → select **Enforced**

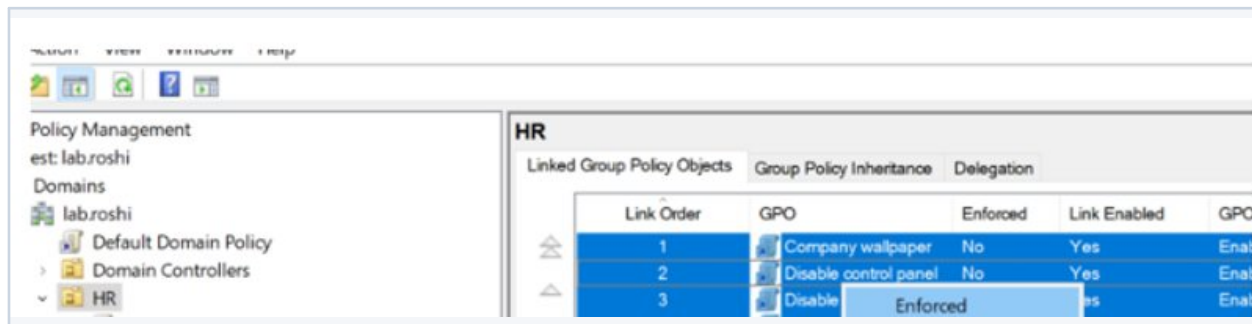


Figure 16 – HR OU showing all three GPOs linked (Company wallpaper, Disable control panel, Disable Task Manager)

4 Confirm the Enforced setting change

- Click **OK** on the confirmation dialog to apply the Enforced setting to each GPO link

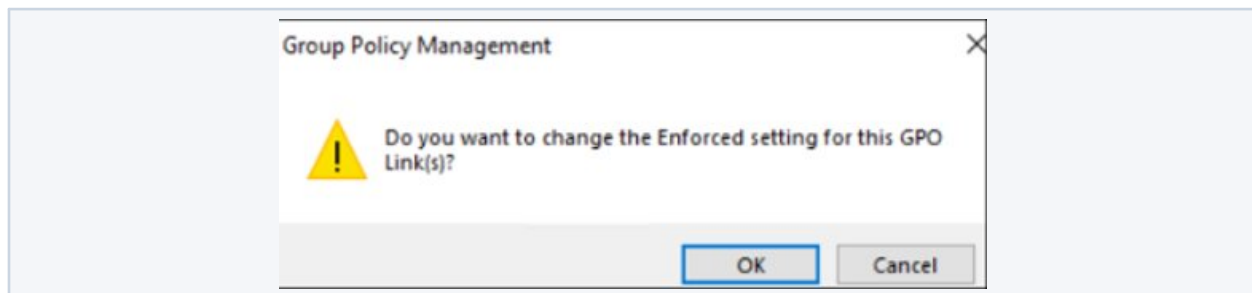


Figure 17 – Confirmation dialog: 'Do you want to change the Enforced setting for this GPO Link(s)?'

GPO Processing Order (LSDOU)

Order	Level	Precedence
L	Local Policy	Applied first — lowest precedence
S	Site Policy	Applied second
D	Domain Policy	Applied third
OU	Organizational Unit Policy	Applied last — highest precedence; overrides all above

7. Applying & Verifying Policies on the Client

After linking and enforcing GPOs, policies apply automatically at the next Group Policy refresh interval (default every 90 minutes). For immediate lab verification, we force a refresh using gpupdate, then log in as domain user mbell to confirm each policy is active.

1 RDP into Client1 as domain user lab.roshi\mbell

- Open Remote Desktop Connection → connect to Client1's IP
- Username: **lab.roshi\mbell** | enter domain user password

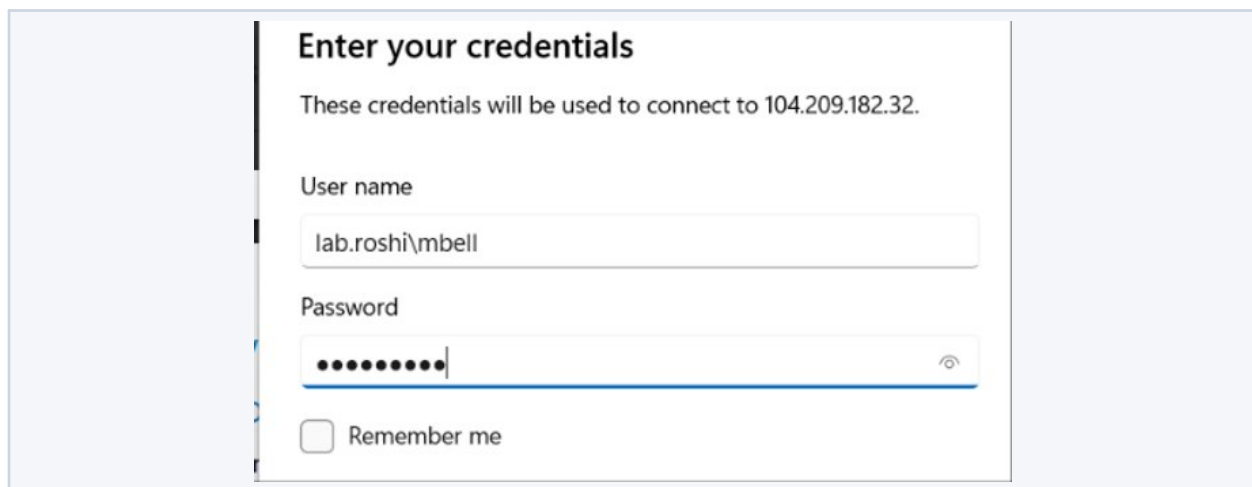


Figure 18 – RDP credential prompt: connecting as lab.roshi\mbell to Client1

2 Force Group Policy refresh on Client1

- Open Command Prompt as Administrator on Client1
- Run:

```
gpupdate /force
```

Verification 1 – Desktop Wallpaper GPO Applied

- After RDP login as mbell, the desktop immediately shows the company wallpaper
- The yellow background with 'Welcome to Roshi Lab!' text and password notice confirms the custom wallpaper GPO is working
- User cannot change the wallpaper via right-click → Personalize

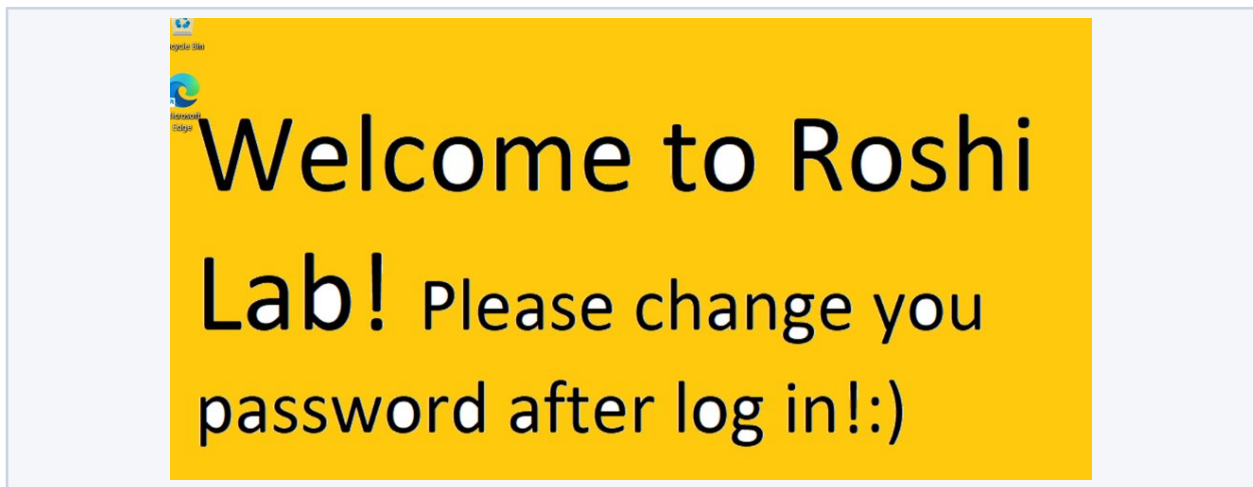


Figure 19 – Client1 desktop showing the company wallpaper (Welcome to Roshi Lab!) enforced by GPO

Verification 2 – Control Panel GPO Applied

- Press **Win + R** → type **control** → press Enter
- Result: Restrictions dialog — 'This operation has been cancelled due to restrictions in effect on this computer'
- Both classic Control Panel and Windows Settings (Win + I) are blocked

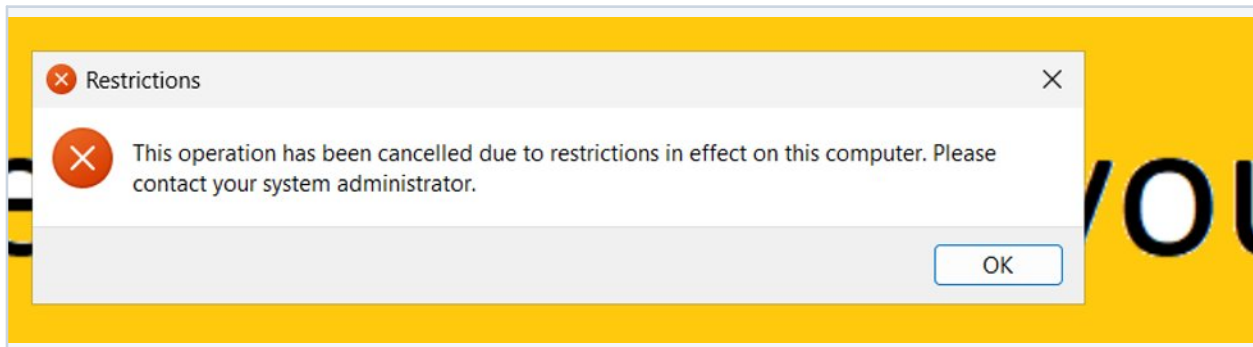


Figure 20 – Restrictions error dialog confirming Control Panel is blocked by GPO on Client1

Verification 3 – Task Manager GPO Applied

- Press **Ctrl + Alt + Del** → Task Manager option is greyed out or absent
- Right-click taskbar → Task Manager → shows 'Task Manager has been disabled by your administrator'

■ All 3 GPOs Verified Desktop Wallpaper, Control Panel restriction, and Task Manager disablement all confirmed active on Client1 after

8. SOC & Identity Security Value

The policies implemented in this lab map directly to enterprise security controls monitored and enforced by Security Operations Centers. The table below summarizes the security value of each GPO from a SOC and identity security perspective.

GPO Policy	Control Type	SOC Relevance	MITRE ATT&CK
Desktop Wallpaper	Compliance / Legal Notice	Legal banner for admissibility; deviation signals unauthorized policy change	T1484 – Group Policy Modification
Disable Control Panel	Least Privilege / Attack Surface Reduction	Prevents network config changes, AV disablement, account manipulation	T1562 – Impair Defenses
Disable Task Manager	Endpoint Hardening / EDR Protection	Prevents killing security agents; monitor re-enable via Event 4657	T1562.001 – Disable Security Tools

Entra ID & Hybrid Identity Context

In modern hybrid environments, on-premises Group Policy works alongside Microsoft Entra ID cloud-based controls. GPOs apply to domain-joined machines on the local network, while Entra ID Conditional Access Policies apply to cloud application access regardless of device location. Together they create a layered identity security posture: GPOs enforce local endpoint hardening while Conditional Access enforces MFA, compliant device checks, and location-based restrictions at the cloud authentication layer.

Control	Scope	Best Used For
Group Policy (GPO)	Domain-joined, on-prem/hybrid	Local OS settings, registry, software restrictions, logon scripts
Entra ID Conditional Access	Cloud/SaaS apps, all devices	MFA enforcement, compliant device checks, risky sign-in blocks
Microsoft Intune (MDM)	All managed devices (cloud-native)	App deployment, compliance policies, replacing GPO in cloud-only

9. Conclusion

This lab successfully demonstrated the end-to-end process of designing, configuring, linking, enforcing, and verifying three Group Policy Objects in a Microsoft Azure-hosted Active Directory environment (lab.roshi). Each policy directly addresses a real enterprise security requirement: enforcing corporate branding and legal notices, restricting access to administrative system tools, and protecting endpoint security agent processes from user-level tampering.

From a security engineering perspective, this lab illustrates how GPOs function as a preventive control layer within the NIST Cybersecurity Framework (Protect function), reducing the attack surface available to both external threat actors and malicious insiders. The policies align with CIS Benchmark controls and DISA STIG requirements for Windows environments.

Key Takeaways

- GPOs are a high-impact, low-cost security control that scale across thousands of endpoints simultaneously.
- Organizational Units (OUs) provide granular targeting — policies can be scoped to specific departments without affecting the entire domain.
- The 'Enforced' flag prevents security baselines from being overridden by local or child-OU policies — essential for compliance.
- **gpupdate /force** is an essential verification tool for confirming policy application during lab and production testing.
- GPOs complement cloud-native Entra ID controls in hybrid environments, forming a comprehensive identity and endpoint security posture.
- SOC teams should monitor Event ID 5136 (AD object modification) and 4657 (registry value change) as indicators of unauthorized GPO tampering — MITRE ATT&CK; T1484.

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